

Enhancing Operational Efficiency and Patient Experience at HealthFirst Care

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Executive Summary

HealthFirst Care's patient-experience initiative, summarized:

- Problem: 41.4-min average wait, 32 double-booking incidents, 26.8% appointment completion rate, and 42.7% of resources unavailable or under maintenance.
- Key insight: manual, paper-based scheduling and check-in — not clinical capacity — drive most delays and errors.
- Key insight: workload and resources are unevenly distributed; Orthopedics runs 7.86 hrs/day against a 3.85-hr average.
- Recommended action: deploy an automated conflict-prevention scheduling engine with live resource-tracking dashboards.
- Recommended action: rebalance staffing to peak-demand windows (Mondays, weekends, 4–6 PM) and rebalance resources toward Pediatrics.
- Expected outcome: wait time ≤ 33.1 min, 0 double-bookings, $\geq 70\%$ resource availability, and satisfaction $\geq 8.0/10$.



Introduction

- Opportunity: rising patient complaints tied to long waits, double-bookings, and uneven resource use.
- Approach: analyze 216 appointment, 217 feedback, 213 resource records; map As-Is/To-Be workflows; design an automated solution.
- Scope: scheduling, check-in, resource allocation, and discharge — stages most tied to dissatisfaction.

- Key question: which workflow stage contributes most to the 41.4-min average wait time?
- Hypothesis: manual, paper-based check-in and scheduling — not clinical capacity — drive most delays.
- Key question: are resources genuinely scarce, or unevenly allocated across departments?



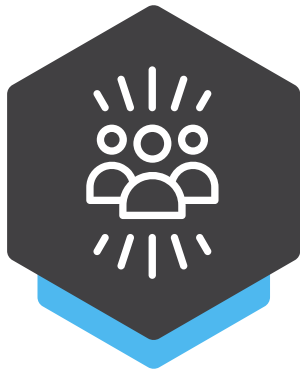
Business Objectives

The management has identified the following areas for improvement:

- Reduce average patient wait time from 41.4 min toward the 33.1-min target (AC-01).
- Eliminate scheduling double-bookings through automated conflict detection (AC-02).
- Raise patient satisfaction from 6.3/10 toward 8.0/10 (AC-03).
- Increase resource availability from 57.3% toward 70% at peak hours (AC-04).
- Rebalance clinical workload — Orthopedics runs 7.86 hrs/day versus a 3.85-hr department average.
- Replace manual, paper-based check-in and billing steps with real-time digital workflows.

Task overview

As a Business Analyst, you have been tasked with driving this initiative. Your role will involve:



Gathering stakeholder requirements



Analyzing existing processes



Developing data-driven solutions to improve operational efficiency



Developing risk assessment and mitigation plan

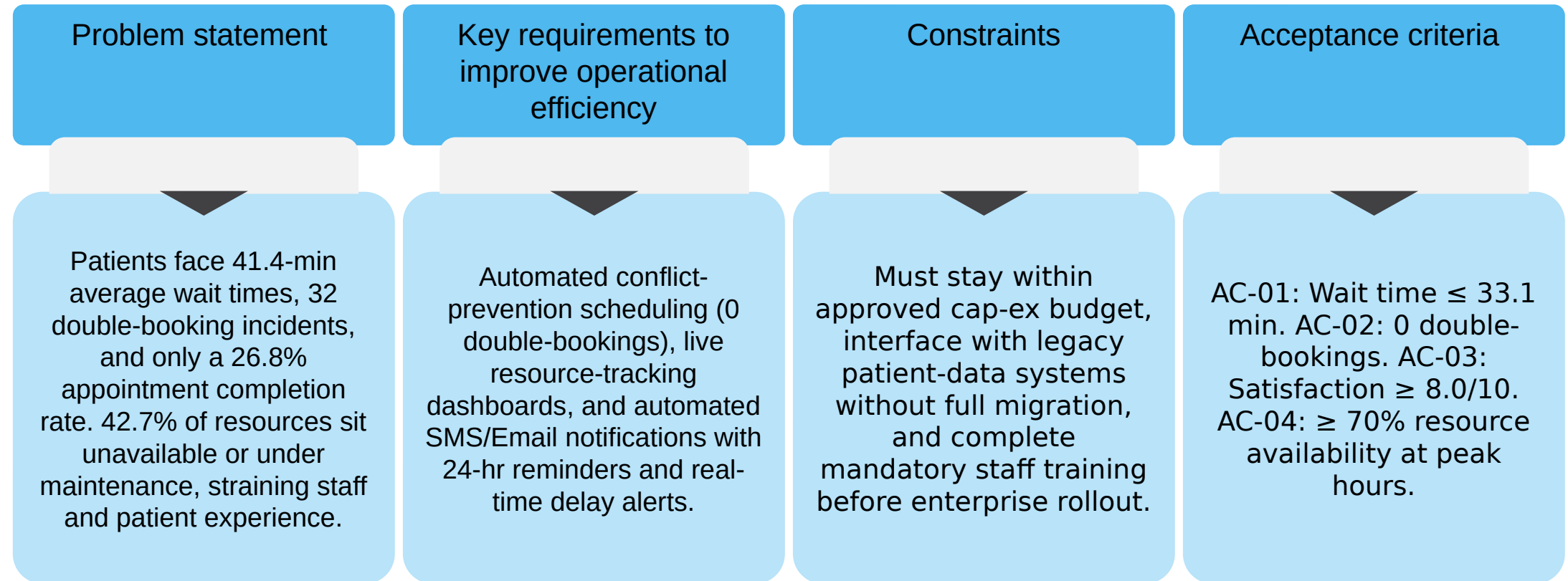


Task 1



Business Requirement Document (BRD)

A BRD defines the business objectives, project scope, key requirements, stakeholder expectations, and deliverables for a project. Your task includes writing the following:



Task 2



Requirement Traceability Matrix (RTM)

A Requirements Traceability Matrix maps and tracks project requirements throughout the lifecycle, ensuring that each requirement is addressed, tested, and aligned with business objectives, stakeholder needs, and project deliverables. You are required to add screenshot of your RTM that:

Categorize requirements into functional and non-functional

Categorize requirements using the MoSCoW method

List requirements based on priority

Requirements Traceability Matrix – HealthFirst Care Initiative

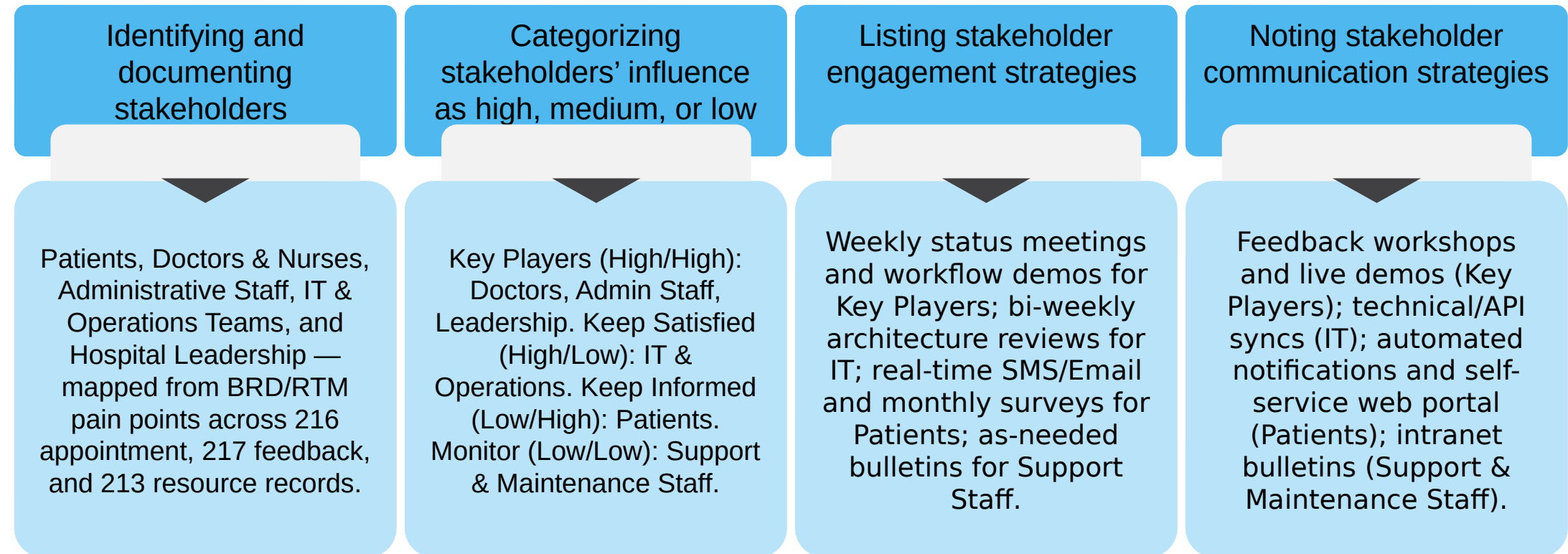
Req. ID	Requirement Description	Priority (MoSCoW)	Stakeholder(s)	Project Objective	Related Data File	Status
FR-01	Automated Conflict Prevention Engine: validate doctor/room/equipment before confirming bookings.	Must Have	Maria Carter, Dr. Khan, Dr. Lee	AC-02: 0 double-booking incidents	appointment_data.csv	Approved
FR-02	Live Resource Tracking: real-time availability dashboard for rooms, equipment, staff.	Must Have	Dr. Lee, Santa Murmu, J. Gomes	AC-04: ≥ 70% resource availability	resource_data.csv	Approved
FR-03	Automated Patient Notification: SMS/Email confirmations, delay alerts, 24-hr reminders.	Must Have	S. Ayvazyan, Lak Ayer, L. Simkow	AC-01/03: wait ≤33.1 min, satisf. ≥8.0	feedback_data.csv	Approved
FR-04	Integrated Patient Record Mgmt: centralized billing, scheduling, and history.	Should Have	Ivan Walker, Rajesh Singh	Eliminate data silos, billing/treatment	feedback_data.csv	Draft
NFR-01	System Scalability & Performance: 99.9% uptime during peak booking windows.	Must Have	Rajesh Singh (IT Manager)	Infrastructure stability at peak load	appointment_data.csv	Approved
NFR-02	Data Security & Compliance: RBAC, encryption, full HIPAA compliance.	Must Have	Laura Simkow, IT & Operations	Address legacy security gaps	N/A	Approved
NFR-03	User-Friendly Interface: simplified multi-generational self-service portal.	Should Have	Lak Ayer, S. Ayvazyan	Reduce booking friction, support-desk volume	feedback_data.csv	Draft

Task 3



Stakeholder Analysis and Engagement Plan

The Stakeholder Analysis and Engagement Plan identifies key stakeholders, understands their interests and influence, and develops strategies to effectively communicate, engage, and manage their expectations throughout the project lifecycle. Your tasks include:



Task 4



Scope Management Plan

The Scope Management Plan defines how the project scope will be planned, documented, validated, and controlled to ensure that all project objectives and deliverables are met while preventing scope creep. Your tasks include:

Categorizing scope into in-scope & out-of-scope

In-scope: automated scheduling engine, resource dashboards, SMS/Email notifications, EHR integration, RBAC security.
Out-of-scope: facility construction, diagnostic hardware, hiring, core billing overhaul (Phase 2).

Identifying assumptions and constraints

Assumes stakeholder availability for bi-weekly reviews and accurate baseline data. Constrained by approved cap-ex budget, HIPAA compliance, and legacy-system interoperability without full migration.

Defining phases in the Work Breakdown Structure (WBS)

WBS — Phases & Milestones

WBS	Task / Work Package	Deliverable / Milestone	Role
1.0	HealthFirst Care Initiative	Enterprise Health System Overhaul	Project Manager
1.1	Requirements & Initiation	Project Charter & Scope Baseline	Business Analyst
1.1.3	Finalize Business Requirements	MILESTONE: Completion of BRD	Project Manager
1.2	System Architecture & Design	Approved Technical Blueprints	Lead IT Architect
1.2.4	Validate Workflows with Stakeholders	MILESTONE: Stakeholder Sign-Off	Clinical Lead
1.3	Software Development & Integration	Fully Integrated Application Suite	Development Team
1.3.2	Build Resource Tracking Dashboard	MILESTONE: Dashboards & Notifications	Senior Developer
1.4	Testing, Validation & Deployment	Production Release & Handover	QA / Ops Team

Note Scope Change Management

Formal Change Request → Impact Analysis (Data Analyst + IT Architect) → CCB Review (PM, IT Architect, Leadership) → Approval & Baseline Update, via a 4-step Change Control Board process.

Task 5



Process Map Diagrams

The Process Map Diagrams visually represent the workflow of a system or process, identifying inefficiencies, redundancies, and areas for improvement to enhance operational efficiency and decision-making. You are required to fill the table with processes along with inefficiencies identified using the As-Is model and areas of improvements identified using the To-Be model:

Process	As-Is model	To-Be model
Appointment Scheduling	Manual availability check; unavailable slots loop patient back with no visibility.	Live slot availability; auto-suggests next best slot.
Check-In & Resource Allocation	Manual paperwork drives 41.4-min avg wait; invalid details loop back to Front Desk.	Kiosk/QR check-in, real-time conflict detection; targets ≤ 33.1 min.
Discharge Planning	No defined billing turnaround; care instructions and billing run sequentially.	Live dashboard, billing SLA with escalation; parallelized nurse/billing steps.

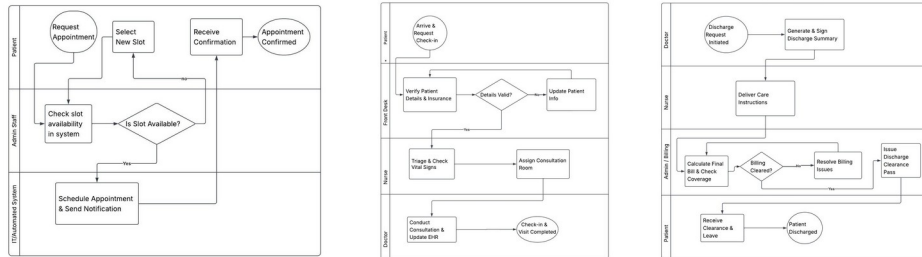
Task 6



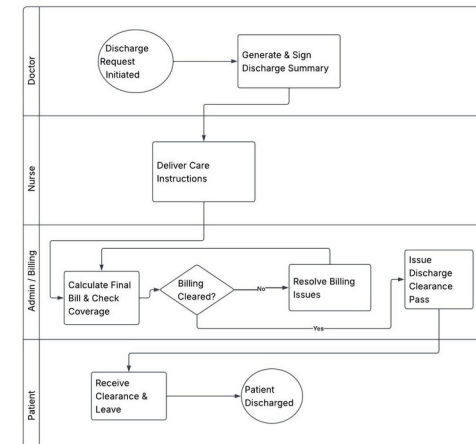
BPMN and Swimlane Diagrams

BPMN and Swimlane diagrams provide a structured visual representation of business processes, clarifying roles, responsibilities, and interactions among different stakeholders to enhance workflow efficiency and communication. Your tasks include adding screenshots for:

Workflow using Advanced BPMN Model



Stakeholder responsibilities using Swimlane diagram



Task 7



Data Analysis Document

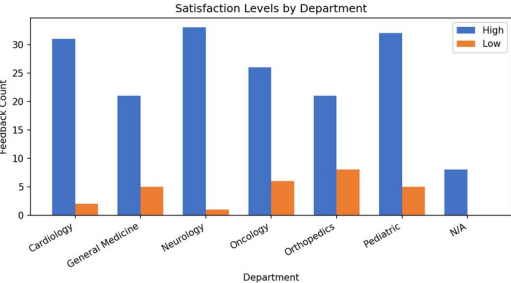
The Data Analysis Document summarizes the key findings, trends, and insights derived from patient and resource data, providing evidence-based recommendations to enhance hospital operations and patient satisfaction. Your tasks include adding screenshots for the following:

Trends identified using
Pivot Table

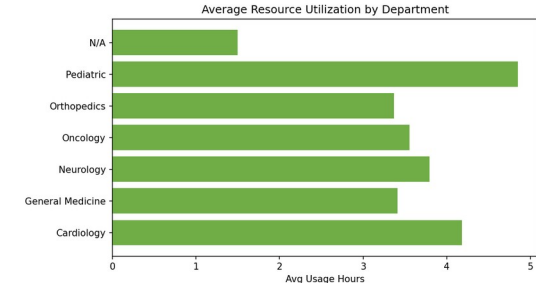
PivotTable — Appointments by Department × Hour of Day

Department	8:00	9:00	10:00	11:00	12:00	13:00	14:00	15:00	16:00	17:00	18:00	Total
Cardiology	5	2	0	3	6	2	4	7	2	4	2	38
General Medicine	3	2	2	2	2	3	2	1	3	2	4	27
Neurology	3	1	5	4	6	0	2	4	2	2	5	35
Oncology	1	3	1	3	2	2	0	3	6	3	5	32
Orthopedics	2	2	1	3	1	2	2	4	2	5	6	30
Pediatric	4	5	5	2	1	3	3	0	3	4	5	37
Grand Total	18	15	14	17	18	12	13	19	18	20	27	199

Analyzed trends



Key insights



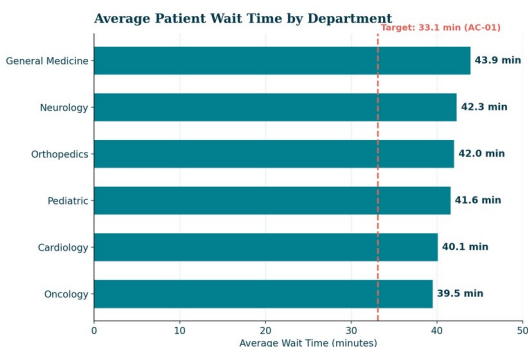
Task 8



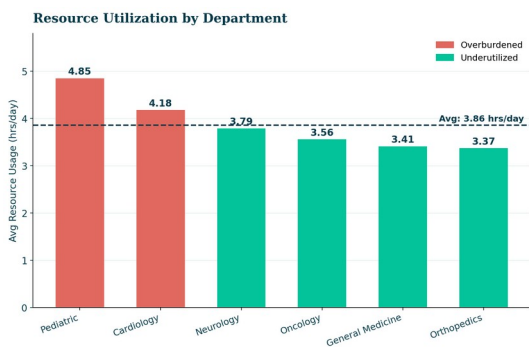
Dashboard

Dashboards visualize key operational metrics, such as patient wait times, resource utilization, and staff productivity, enabling data-driven decision-making for improved hospital efficiency and patient experience. Your task is to add the following screenshots:

Average patient wait time using a Bar Chart

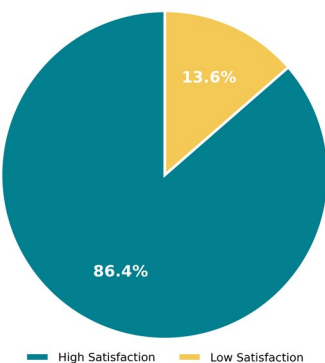


Bar Chart highlighting overused and underutilized resources

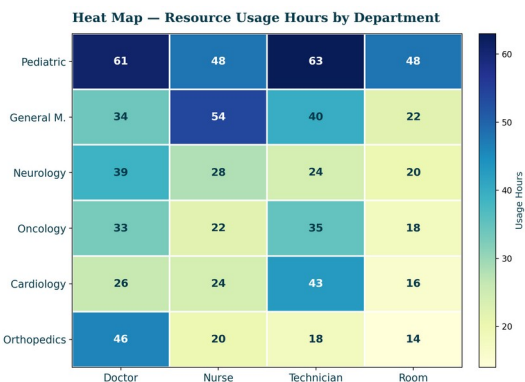


Patient feedback visualized using a Pie Chart

Patient Feedback — Satisfaction Level



Heat Map showing the efficiencies of the departments



Task 9



Risk Management Plan

The Risk Management Plan identifies, assesses, and mitigates potential risks—such as data privacy concerns, system integration challenges, and operational inefficiencies—to ensure smooth project execution and improved patient care. Your tasks include adding screenshots from your Risk Management plan based on the specified requirements:

Risks identified in the risk register

Categorized risks based on the Risk Assessment Matrix

Elements identified in the SWOT Analysis

Key insights

HealthFirst Care Initiative — Risk Register

Project Governance & Operational Efficiency Dashboard (Prepared by: PMO-001)

Risk ID	Risk Description	Risk Category	Likelihood
CR-01	Automated workflow integration for patient conflict prevention engine, ensuring manual interventions are minimized before go-live.	Operational	Medium
OP-02	Mandatory staff training on legacy and the new scheduling platform to minimize errors during go-live.	Operational	Medium
OP-03	Implementing a contingency plan for potential system downtime during the transition period.	Operational	High
OP-04	Performing a full-scale integration test, including data migration and system validation, before the go-live.	Operational	High
OP-05	Secure network infrastructure (firewalls, intrusion detection) to protect patient data and system integrity.	Operational	Medium
TE-01	Cloud-enabled EHR integration with legacy patient data architecture, ensuring data integrity and security.	Technical	Low
TE-02	The integration of patient appointment scheduling system, ensuring the EHR system is fully functional.	Technical	Medium
TE-03	Implementing a robust backup and recovery strategy for patient data, ensuring data availability.	Technical	Medium
TE-04	Ensuring the EHR system is fully functional and integrated with the legacy system, ensuring data integrity.	Technical	Medium
TE-05	Self-service kiosks or QR codes in hardware, ensuring a robust and secure system.	Technical	Low
ST-01	Stakeholder engagement and communication plan, ensuring all stakeholders are informed and aligned.	Stakeholder	Medium
ST-02	Stakeholder engagement and communication plan, ensuring all stakeholders are informed and aligned.	Stakeholder	Medium
ST-03	Low engagement from Support & Maintenance staff, ensuring a robust and secure system.	Stakeholder	Low
ST-04	Stakeholder engagement and communication plan, ensuring all stakeholders are informed and aligned.	Stakeholder	Medium
ST-05	Stakeholder engagement and communication plan, ensuring all stakeholders are informed and aligned.	Stakeholder	Medium

Impact	Severity Score	Severity Level	Mitigation Strategy
High	6	High	Proactive review with a patient care period, structured change management plan with weekly training sessions for Maria Carter (Senior Governance Officer).
Medium	4	Medium	Proactive review with a patient care period, structured change management plan with weekly training sessions for Maria Carter (Senior Governance Officer).
Medium	6	High	Define a service level target (e.g., 2-hour resolution) for all service requests, ensuring a dedicated support desk for the first two weeks post-launch.
High	9	High	Implement a contingency plan for potential system downtime during the transition period.
High	6	High	Perform a full-scale integration test, including data migration and system validation, before the go-live.
High	3	Medium	Secure network infrastructure (firewalls, intrusion detection) to protect patient data and system integrity.
High	6	High	Cloud-enabled EHR integration with legacy patient data architecture, ensuring data integrity and security.
High	6	High	The integration of patient appointment scheduling system, ensuring the EHR system is fully functional.
Medium	4	Medium	Implementing a robust backup and recovery strategy for patient data, ensuring data availability.
High	6	High	Ensuring the EHR system is fully functional and integrated with the legacy system, ensuring data integrity.
Medium	2	Low	Self-service kiosks or QR codes in hardware, ensuring a robust and secure system.
Medium	4	Medium	Stakeholder engagement and communication plan, ensuring all stakeholders are informed and aligned.
Medium	4	Medium	Stakeholder engagement and communication plan, ensuring all stakeholders are informed and aligned.
Medium	4	Medium	Stakeholder engagement and communication plan, ensuring all stakeholders are informed and aligned.
Medium	2	Low	Low engagement from Support & Maintenance staff, ensuring a robust and secure system.
Medium	4	Medium	Stakeholder engagement and communication plan, ensuring all stakeholders are informed and aligned.
High	6	High	Stakeholder engagement and communication plan, ensuring all stakeholders are informed and aligned.

7 High-severity risks (score 6-9) concentrate in Operational execution and system reliability; 6 Medium risks are tracked via routine review; 2 Low risks (TE-05, ST-03) need only periodic checks.

Strengths: strong sponsorship, validated baseline data. Weaknesses: manual billing exceptions, legacy silos. Opportunities: automated conflict detection, resource rebalancing. Threats: HIPAA risk, stakeholder resistance.

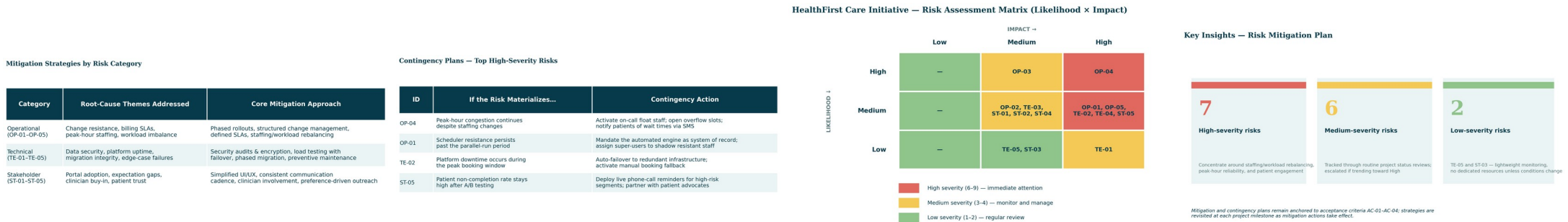
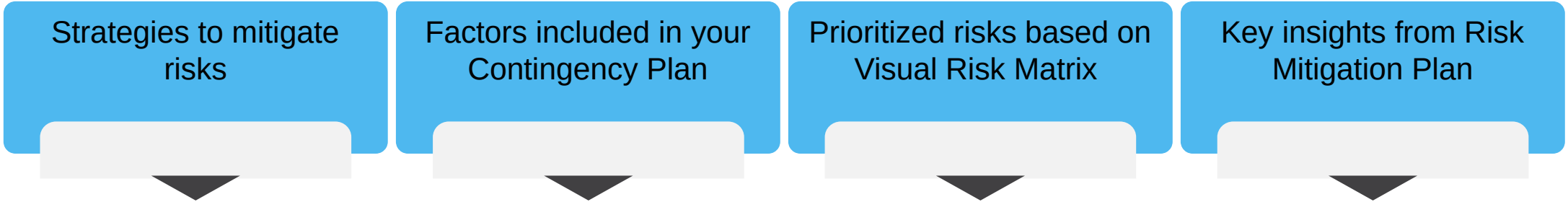
High-severity risks cluster around staffing/workload rebalancing, peak-hour reliability, and patient engagement — the same themes surfaced across the BRD, process models, and swimlane analysis.

Task 10



Risk Mitigation Plan

The Risk Mitigation Plan identifies potential risks, assesses their impact, and implements strategies to minimize disruptions, ensuring smooth hospital operations and improved patient care. Your task includes adding screenshots from your Risk Mitigation plan based on the specified requirements:





Key Findings

Three findings emerged consistently across the process, data, and risk analysis:

- Finding 1 — Manual processes are the primary bottleneck: front-desk paperwork and manual availability checks drive the 41.4-min average wait and all 32 double-booking incidents, not clinical capacity.
- Finding 2 — Resource and workload distribution is uneven, not scarce: Pediatric resources run at 4.85 hrs/day while General Medicine and Orthopedics sit near 3.4 hrs/day; Orthopedics clinicians average 7.86 hrs/day.
- Finding 3 — Dissatisfaction concentrates at predictable peaks: Mondays, weekends, and the 6 PM hour see both the highest appointment volume and the lowest satisfaction scores.



Key Recommendations

Four actionable recommendations for HealthFirst Care leadership:

- Deploy the automated conflict-prevention scheduling engine and live resource dashboard to eliminate double-bookings and give staff real-time visibility.
- Rebalance peak-hour staffing toward Mondays, weekends, and the 4–6 PM window, where volume and dissatisfaction peak together.
- Rebalance resources and workload from underutilized General Medicine/Orthopedics toward the overburdened Pediatric department.
- Enforce the risk mitigation cadence — weekly review of the 7 high-severity risks, with defined SLAs for billing and platform uptime.



Conclusion

Summary of observations:

- HealthFirst Care's patient-experience challenges stem primarily from manual, paper-based processes rather than a genuine shortage of clinical capacity.
- The proposed automated scheduling, resource-tracking, and notification solution directly targets the root causes identified in the As-Is process and data analysis.
- Success depends on disciplined change management — phased rollout, staff training, and active risk mitigation — alongside the technical solution itself.
- Meeting the AC-01–AC-04 acceptance criteria (wait time, double-bookings, satisfaction, resource availability) would materially improve both patient experience and operational efficiency.